

A yellow L-shaped graphic element is positioned to the left of the main title.

WELCOME TO APPLIED GIS FUNDAMENTALS!

A yellow L-shaped graphic element is positioned to the right of the main title.

Valdesera Christina Lydia
christina.valdesera@uni-graz.at

07 October 2026

Welcome

Instructor



Valdesera Christina Lydia (PhD student at the Geospatial Technologies team)

christina.valdesera@uni-graz.at

Office hours: By appointment

Agenda

01 Welcome

02 Get to know each other

03 Course structure

04 Learning outcomes

05 Grading criteria & expectations

06 AI & GIS
Opportunities-Challenges

07 GitHub



08 QGIS installation

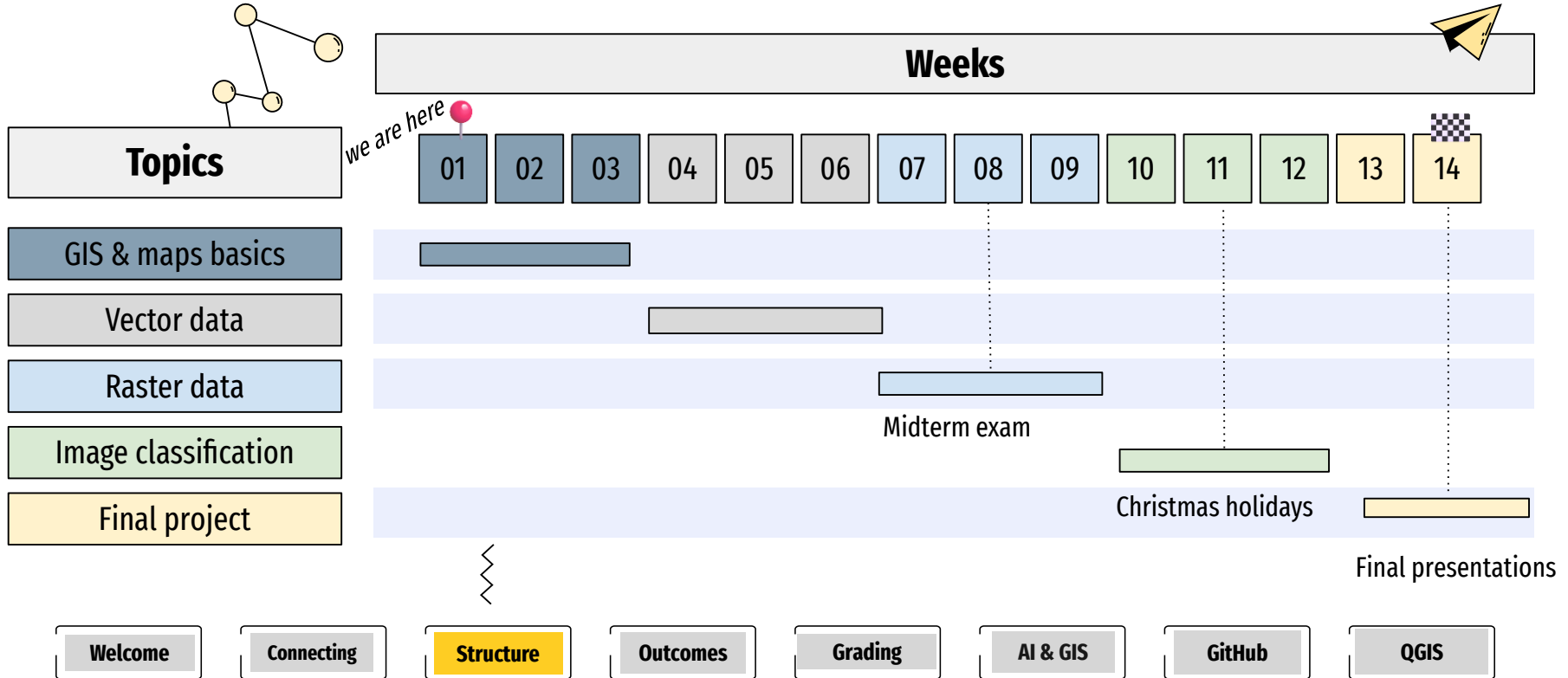


Get to know each other

Scan me and let's discuss!



Course structure & timeline



Learning outcomes

- Use QGIS software to manage, visualize and analyze spatial data
- Apply georeferencing techniques, work with vector and raster data and conduct basic image classification
- Create maps and spatial analysis outputs
- Collaborate in groups to design and execute GIS projects

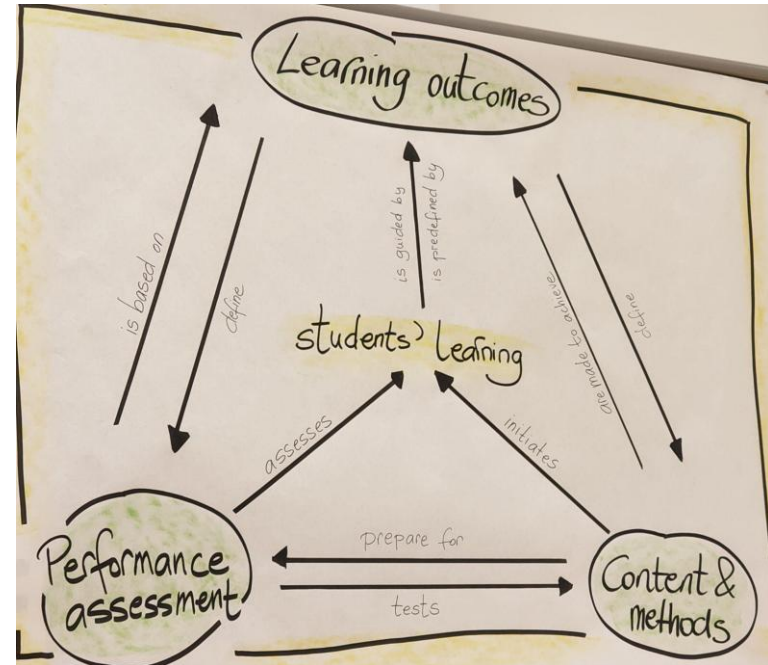


chart by MMag. Dr. Elisabeth Weber

Grading criteria

Grading Scheme

Final group project 45 %

Weekly lab assignments 45 %

Participation 5 %

Midterm exam 5 %



Expectations

- Bring your own laptop
- Download the data before the weekly lab
- Weekly attendance
- Submission of assignments on time (sent by email)

Email subject format: “402.467_LabX_LastName_FirstName”

- Equal contribution in group projects
- Active participation

Why GIS matters

GIS is being used to answer questions such as:

- Where is it? 
- What else is nearby?
- Where is the highest concentration of 'X'?
- Where can I find things with characteristic 'Y'?
- Where is the closest 'Z' to my location?

Tobler's first law of Geography

"Everything is related to everything else, but near things are more related than distant things"

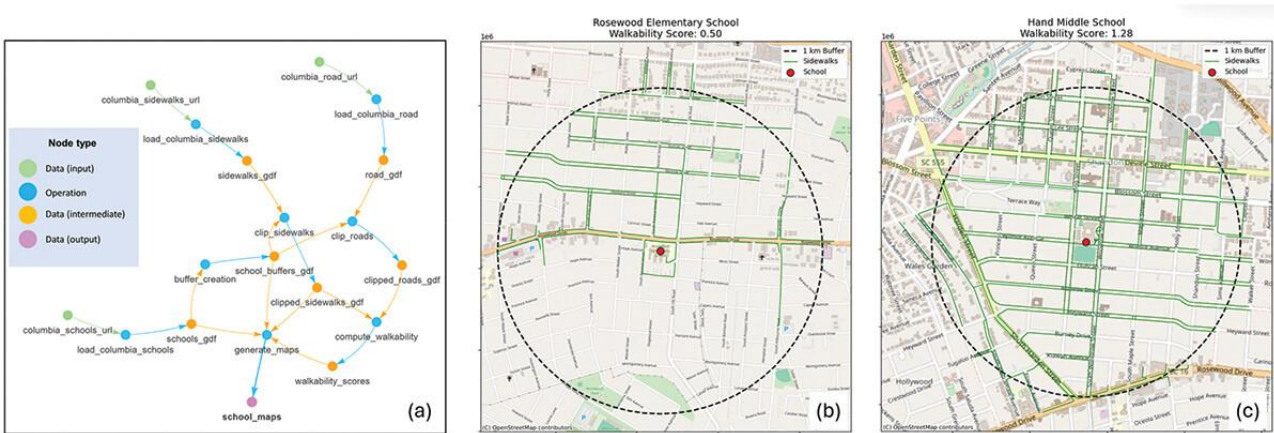
GIS example



@Stadt Graz

GIScience in the era of Artificial Intelligence: a research agenda towards Autonomous GIS

Zhenlong Li^{a*}, Huan Ning^{a*}, Song Gao^b, Krzysztof Janowicz^c, Wenwen Li^d, Samantha T. Arundel^e, Chaowei Yang^f, Budhendra Bhaduri^g, Shaowen Wang^h, A-Xing Zhu^b, Mark Gaheganⁱ, Shashi Shekhar^j, Xinyue Ye^k, Grant McKenzie^l, Guido Cervone^m and Michael E. Hodgsonⁿ

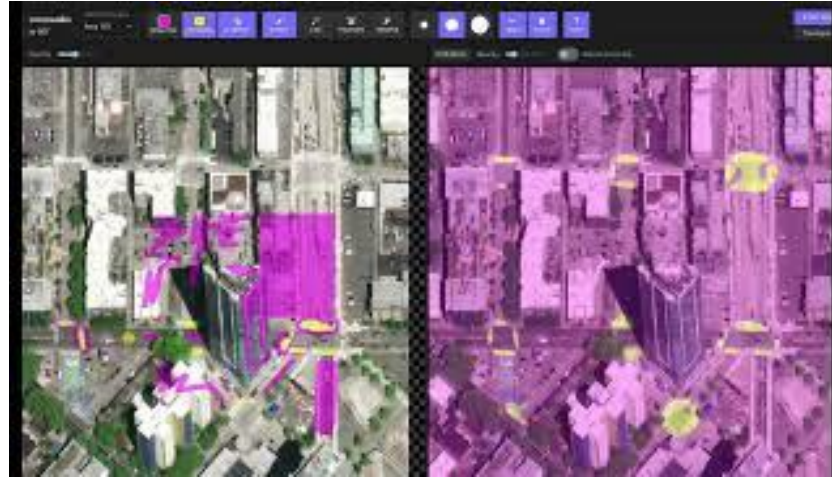


The results of the school walkability assessment by LLM-Geo.

(a) The generated workflow. (b), (c) Two school map examples that show the walkability score, buffer, and sidewalks.

AI in GIS

Opportunities



@Blackshark.ai

Deep fake Geography



Cartography and Geographic Information Science



ISSN: (Print) (Online) journal homepage: <https://www.tandfonline.com/loi/tcag20>



Frontiers in Big Data

TYPE Review
PUBLISHED 04 September 2024
DOI: 10.3389/fdata.2024.1400024



OPEN ACCESS

EDITED BY
Nikolaos Pitropakis,
Edinburgh Napier University, United Kingdom

REVIEWED BY
Riccardo Spolaor,

Deepfake: definitions,
performance metrics and
standards, datasets, and a
meta-review

Deepfake Geography: Detecting AI-Generated Satellite
Images

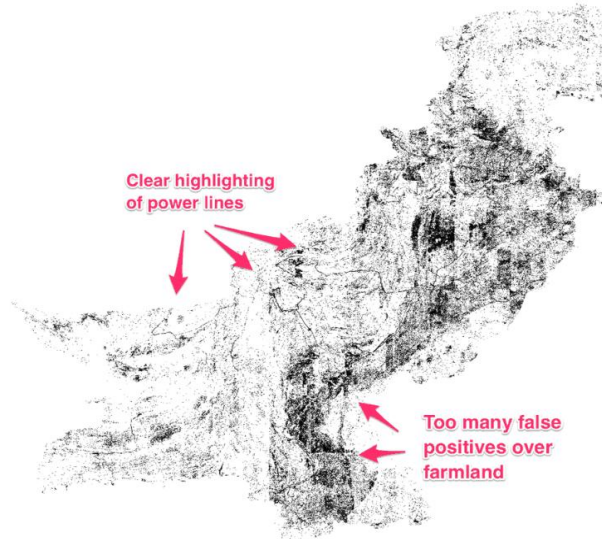
Mansur Yerzhanuly



Critically evaluate AI results!

AI in GIS

Errors in feature detection



@GeographyRealm

Challenges

Welcome

Connecting

Structure

Outcomes

Grading

AI & GIS

GitHub

QGIS

GitHub



GitHub repository: https://github.com/christinavaldesera-gif/402.467_Applied-GIS-Fundamentals.git

Discussions section. Remember we talked about active participation? **This is also the space to do so!**

GitHub

Note

Highlights information that users should take into account, even when skimming.

Tip

Optional information to help a user be more successful.

Important

Crucial information necessary for users to succeed.

Warning

Critical content demanding immediate user attention due to potential risks.

Caution

Negative potential consequences of an action.

1 Download the software

- You can download the latest version here: <https://www.qgis.org/download/>

Recommendation: download the latest Long Term Release version as it is more stable

2 Depending on your operating system, choose between *Download for Windows*, *Download for macOS* or *Download for Linux*

If you are stuck, feel free to check:

- QGIS official Installation guide: <https://qgis.org/resources/installation-guide/>
- YouTube tutorials on *How to install QGIS*: <https://www.youtube.com/watch?v=88rUyahYrDk>

Note! Set the interface language to **English**

QGIS interface

Check official QGIS documentation: [2.1. Lesson: An Overview of the Interface](#)

